



Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is the function of the Instruction Register (IR)?
- (ii) What is the difference between SRAM and DRAM?
- (iii) What is the role of the MAR (Memory Address Register)?
- (iv) What is the difference between RISC and CISC architectures?
- (v) What is the purpose of a Bus in computer architecture?
- (vi) What is the function of a Decoder in a CPU?
- (vii) Explain the difference between I/O-mapped I/O and memory-mapped I/O.
- (viii) What is the purpose of an interrupt vector table?
- (ix) What is instruction-level parallelism (ILP)?
- (x) What is the role of a shadow register in computer architecture?
- (xi) What does the term 'speculative execution' refer to in modern processors?
- (xii) Which type of parallelism does SIMD (Single Instruction, Multiple Data) exploit?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What are the main differences between primary, secondary, and cache memory? [5]
3. Define pipelining and explain its stages. [5]
4. Differentiate between hardwired control and microprogrammed control. [5]
5. Describe the working of an arithmetic logic unit (ALU) with a diagram. [5]
6. Explain the concept of Register Renaming and how it mitigates Write-After-Write (WAW) and Write-After-Read (WAR) hazards in modern processors. Provide a practical example. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Explain the concept of virtual memory and its advantages. [8]
(b) What are the different types of addressing modes in computer architecture? Provide examples. [7]
8. (a) Explain the concept of instruction cycle and its stages. [8]
(b) Describe the function of the control unit in a CPU. [7]
9. (a) Discuss the role of cache coherence in multiprocessor systems. [5]
(b) Describe the key functions of device controllers in computer systems. [10]
10. (a) Discuss the importance of performance metrics such as MIPS and FLOPS. [5]
(b) Explain the role of DMA (Direct Memory Access) in improving system [10]
11. (a) Describe the architecture and working of RAID (Redundant Array of Independent Disks) levels. [8]
Discuss the pros and cons of different RAID levels.
(b) Explain the concept of Data Forwarding in pipelined processors and how it resolves data hazards. [7]
Provide a practical example.

*** END OF PAPER ***